PAPERGRAPH / CRISPR OBSERVABILITY

Follow what is observed. Keep every inference conditional.

Question → contract → evidence → conditional conclusion → frontier.

Read stations 01–12. Dotted arrows mean “read next”, not “proves”.

01 QUESTION

What can the readout justify?

Bind every percentage to an observation contract.

429,812 records / 15 streams; primary 177,006 / 9.

Methods: units; Table S1

02 DEFINITION

Declare event and denominator

Local DNA, exact coding and software modification differ.

Local peptide ≠ exact DNA ≠ MODIFIED/aligned.

Methods: events; Tables 4A–4B

04 FORMAL DERIVATION

Retain unknowns; derive bounds

Finite bounds require stated label and completion premises.

N 20,573 · E 3,563 · NE 15,846 · U 1,164

Table 2; Propositions 2–3

03 MECHANISM

The product can hide its anchor

Same reads; native PE−mock contrast under common Q30.

-10.09177 / +4.38081 / +13.23196 pp

CFB-26-16 lower: edge/inner/target

Table 1; Figure 1; Prop. 1

05 COUNTEREXAMPLE

Equal labels, different strings

Compare observed coding strings before joining outputs.

170,808 joint; 83 unequal strings; 18 retain equal native-event labels.

Results: correspondence; Figure 4

06 METHOD

Require a common observed string

Stricter correspondence retains unresolved mass.

170,725 / 177,006 retained; 6,281 U.

Target + two baseline projections.

Tables 2, S7; Proposition 4

08 SENSITIVITY

Stable sign, changing labels

Contrast stability does not imply record-label stability.

144 jobs / 14 OAT profiles; 163 native conflicts; stock separate.

Table S12; finite-grid Results

07 COMPUTATIONAL RESULT

Read the conditional result

Native PE−mock positivity requires the accepted-label premise.

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Table S8; Proposition 5

09 FORMAL DERIVATION

Preserve a declared target

Check fibers and joint logic on resolved records.

Resolved: 12,852 / 128 / 157,745 (exact+native/native-only/neither)

Supplement: joint targets / criterion

10 INTERPRETIVE CASES

Keep the counterexamples

Template ambiguity and constant targets limit the certificate.

P23H/WT: 572 U assignment weight

EMX1: two constant targets

Table 3; ambiguity / constant targets

12 CONSEQUENCE AND FRONTIER

Report the result and frontier

State sources, conditions and what would revise the claim.

36 cards / 24 compatible contrasts.

Case-tested reporting proposal.

Methods: contract; revision tests

11 REPRODUCIBILITY

Separate replay and independence

Shared sources define what computation can verify.

177,006 raw / 354,012 per-ref labels

Shared-source alignment replay

Methods: verification; replay levels

A FORMAL DEPENDENCY, NOT A CAUSAL ARROW

Nonnegative integer partition
E + NE + U = N; N > 0
Correct labels; unknown labels free

AND

Attainable: p = (E+k)/N
integer k = 0,…,U
Hull: [E/N, (E+U)/N]

Conditional theorem: not sampling coverage or molecular truth.

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